

Problem Set on Sequences

Problem 1. Show that the sequence

$$u_n = \left(1 + \frac{1}{n}\right)^n$$

is monotone increasing and bounded above by 2.

Problem 2. Show that the sequence

$$v_n = \left(1 + \frac{1}{n}\right)^{n+1}$$

is monotone decreasing and $\lim_{n \rightarrow \infty} v_n = e$.

Problem 3. (i) Give an example of a sequence of rational numbers that converges to an irrational number.

(ii) Give an example of a sequence of irrational numbers that converges to a rational number.

(iii) Give an example of divergent sequences $\{u_n\}$ and $\{v_n\}$ such that the sequence $\{u_n + v_n\}$ is convergent.

(iv) Give an example of divergent sequences $\{u_n\}$ and $\{v_n\}$ such that the sequence $\{u_n v_n\}$ is convergent.

Problem 4. Let $\{u_n\}$ be a bounded sequence and $\lim v_n = 0$. Prove that $\lim u_n v_n = 0$.

Utilise this to prove that

$$(i) \lim \frac{a_n}{n} = 0, \quad (ii) \lim \frac{(-1)^n n}{n^2 + 1} = 0, \quad (iii) \lim (-1)^n a_n = 0 \text{ if } \lim a_n = 0.$$

Problem 5. Let $\{u_n\}, \{v_n\}$ be two real sequences with $\lim u_n = l$, $\lim v_n = m$. If $x_n = \max\{u_n, v_n\}$, $y_n = \min\{u_n, v_n\}$ prove that the sequence $\{x_n\}$ converges to $\max\{l, m\}$ and the sequence $\{y_n\}$ converges to $\min\{l, m\}$.

[Hint: $\max\{a, b\} = \frac{1}{2}(a + b + |a - b|)$, $\min\{a, b\} = \frac{1}{2}(a + b - |a - b|)$ for all $a, b \in \mathbb{R}$.]

Problem 6. Prove that the sequence $\{u_n\}$ is a null sequence.

$$(i) u_n = \frac{n!}{n^n}, \quad (ii) u_n = \frac{3^n}{4^{3n}}, \quad (iii) u_n = \frac{b^n}{n^p}, \quad b > 1.$$

Problem 7. 7. Use Sandwich Theorem to prove that

$$(i) \lim(\sqrt{n+1} - \sqrt{n}) = 0, \quad (ii) \lim \frac{(2^n + 3^n)}{n^n} = 3,$$

$$(iii) \lim \left(\frac{1}{n+1} + \frac{1}{n+2} + \cdots + \frac{1}{2n} \right) = 0, \quad (iv) \lim \frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{2 \cdot 4 \cdot 6 \cdots (2n)} = 0.$$

Problem 8. Establish the convergence and find the limit of the sequence $\{u_n\}$, where

$$u_n \text{ is } (i) \left(1 + \frac{1}{3n}\right)^n, \quad (ii) \left(1 + \frac{1}{n^2}\right)^{n^2}, \quad (iii) \left(1 + \frac{1}{3n+1}\right)^n, \quad (iv) \left(1 + \frac{1}{n^2+2}\right)^{n^2}.$$

Problem 9. Prove that the sequence $\{u_n\}$ is convergent by showing that the subsequences $\{u_{2n}\}$ and $\{u_{2n-1}\}$ converge to the same limit.

$$(i) \quad 0 < u_1 < u_2 \text{ and } u_{n+2} = \frac{1}{3}(u_{n+1} + 2u_n) \text{ for } n \geq 1;$$

$$(ii) \quad 0 < u_1 < u_2 \text{ and } u_{n+2} = \sqrt{u_{n+1}u_n} \text{ for } n \geq 1;$$

$$(iii) \quad 0 < u_1 < u_2 \text{ and } \frac{2}{u_{n+2}} = \frac{1}{u_{n+1}} + \frac{1}{u_n} \text{ for } n \geq 1.$$

Problem 10. Prove that the sequence

$$(i) \left\{ 2, \frac{2}{1+2}, \frac{2}{1+\frac{2}{1+2}}, \dots \right\} \text{ converges to } 1; \quad (ii) \left\{ 6, \frac{6}{1+6}, \frac{6}{1+\frac{6}{1+6}}, \dots \right\} \text{ converges to } 2.$$

Problem 11. $\{x_n\}$ and $\{y_n\}$ are bounded sequences and a sequence $\{z_n\}$ is defined by

$$z_1 = x_1, \quad z_2 = y_1; \quad z_3 = x_2, \quad z_4 = y_2; \quad z_5 = x_3, \quad z_6 = y_3; \quad \dots$$

Prove that the sequence $\{z_n\}$ is convergent if and only if both the sequences $\{x_n\}$ and $\{y_n\}$ are convergent with the same limit.